



Data Céilí 2025

Hands on with Fabric: end-to-end data engineering



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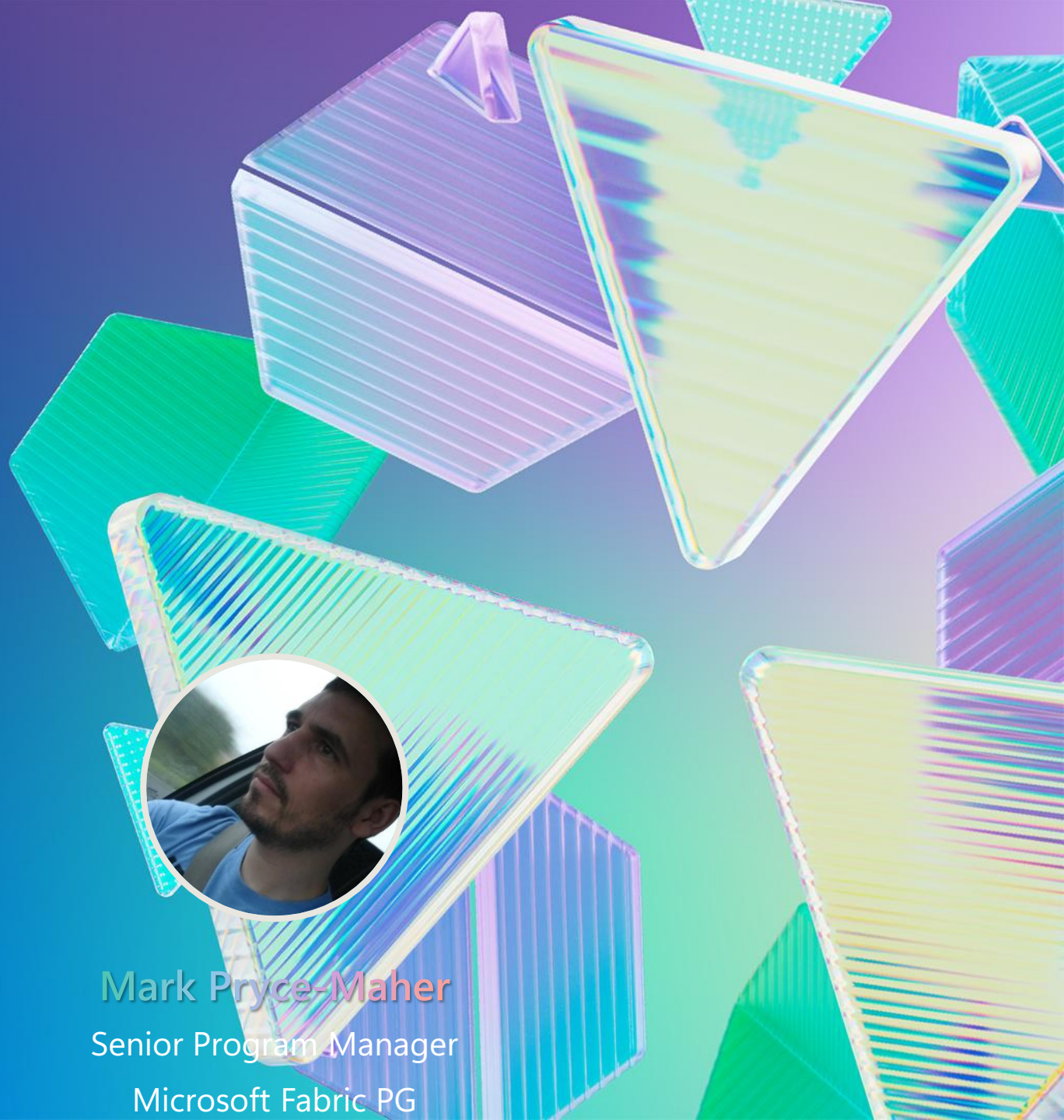
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Introduction, setup and overview of Fabric



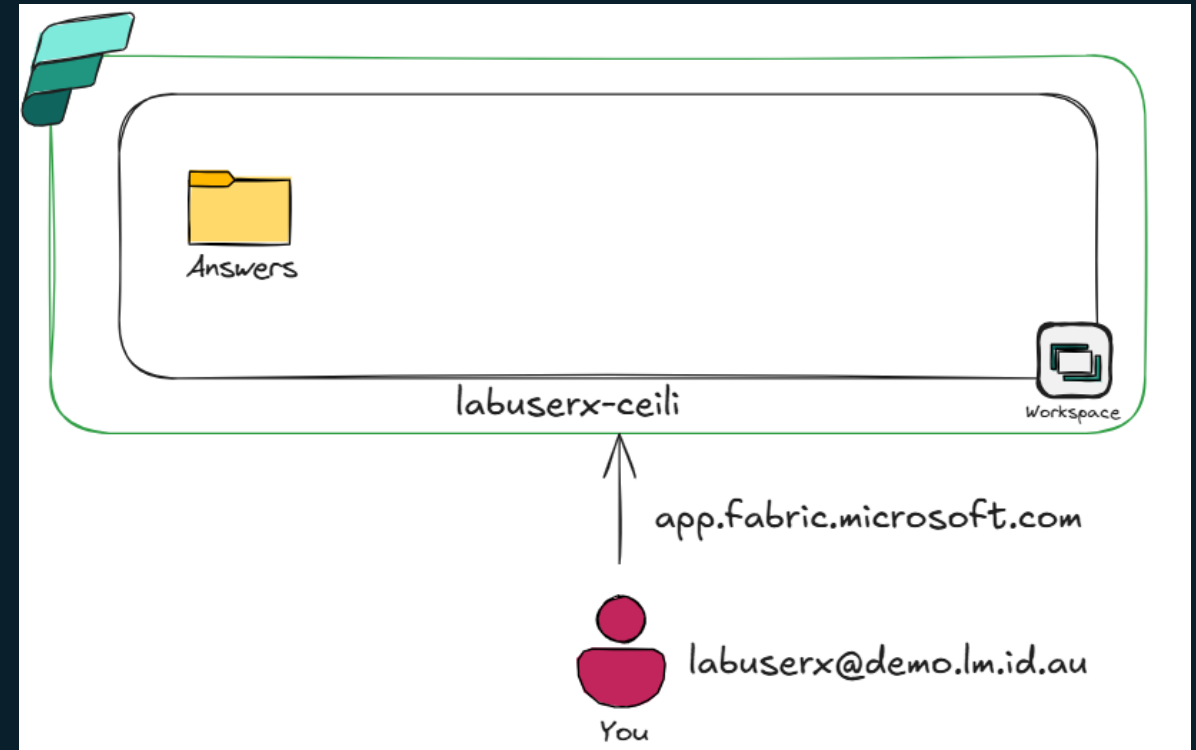
Agenda

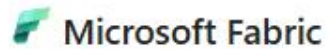
09:00 am – 09:30 am	Introduction, setup and overview of Fabric Analytics platform
09:30 am – 10:45 am	Ingest data using Fabric Data Factory, Copy Job and Mirroring
10:45 am – 11:00 am	Break
11:00 am – 12:00 pm	Transform data using Spark and SQL
12:00 pm – 01:00 pm	Lunch break
01:00 pm – 02:00 pm	Query data using Fabric Data Agents and interact with OneLake
02:00 pm – 02:15 pm	Break
02:15 pm – 03:45 pm	Monitoring, admin, capacities and automation
03:45 pm – 04:00 pm	Break
04:00 pm – 05:00 pm	Wrap up and Q&A



Setup

- Each participant will get a **unique user ID** and **Fabric Workspace**
 - User ID (no need to set up MFA): used to log in to Fabric - <https://app.fabric.microsoft.com>
 - Workspace: used to deploy artifacts and run code
- The **Fabric workspace is all you need** – no need to install any local software or use other tooling
- Your Fabric workspace contains a folder named “answers” which contains all the artifacts & code you need in case you’re stuck
- We will make it clear what you need to deploy yourself vs what will be provided by us





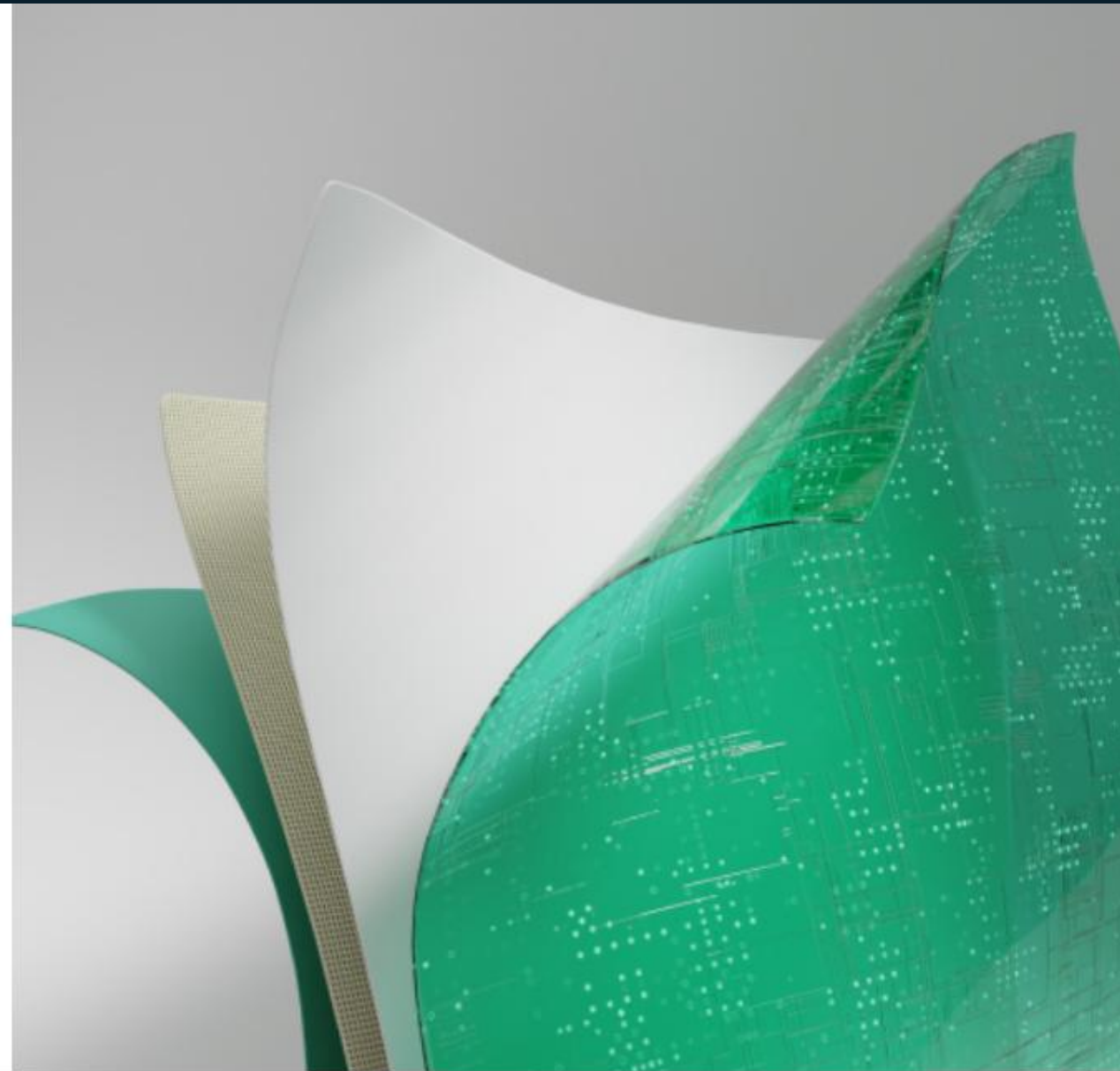
Enter your email, we'll check if you need
to create a new account.

Email

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Submit



Scenario

In a bold quest to uncover the **true source of happiness**, you decide to analyse whether countries with top-tier beer are generally more cheerful than those stuck with sad, watery brews. But just to be thorough, you also want to know if sheer volume—i.e., drinking *a lot* of mediocre beer—can make up for the lack of quality. It's science, really: hops vs. hope. The fate of global joy may hinge on one (or a lot of) pints.



General rules & guidelines

1. We're all here to learn – ask questions!
2. Friendly environment – there are no "stupid" questions
3. In case you're stuck or find it hard to follow, please bring it to our attention
4. Be respectful at all times and if you can, help your neighbour
5. Don't use your Fabric Workspace for crypto mining, DDoS attacks or other illegal activities
6. Have fun!

RULES & GUIDELINES



Respect



Ask Questions



Help Each Other



No Illegal Activities



Have Fun



Microsoft Fabric

The unified data platform for AI transformation



Data
Factory



Analytics



Databases



Real-Time
Intelligence



Power BI



Industry
Solutions



Partner
Workloads



AI



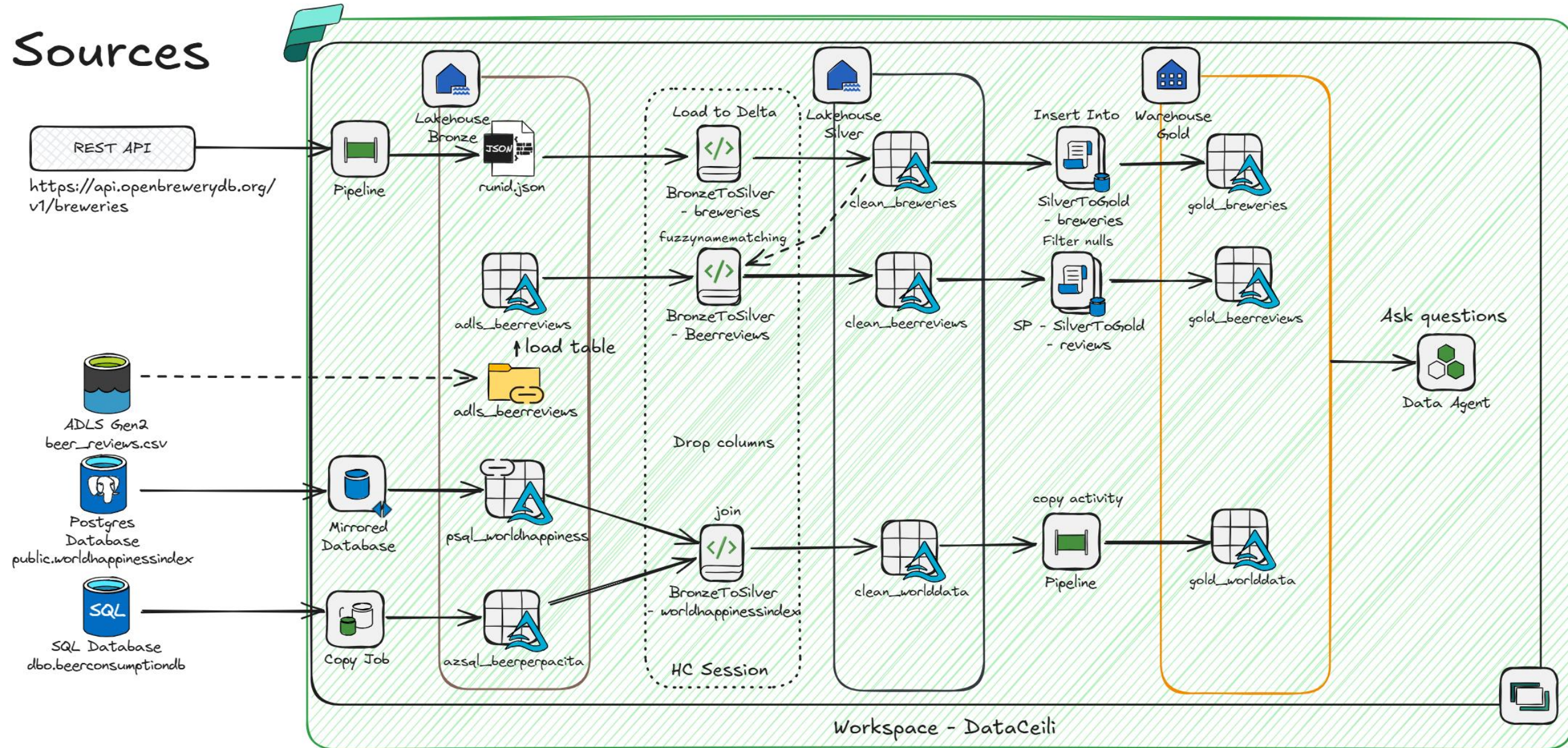
OneLake



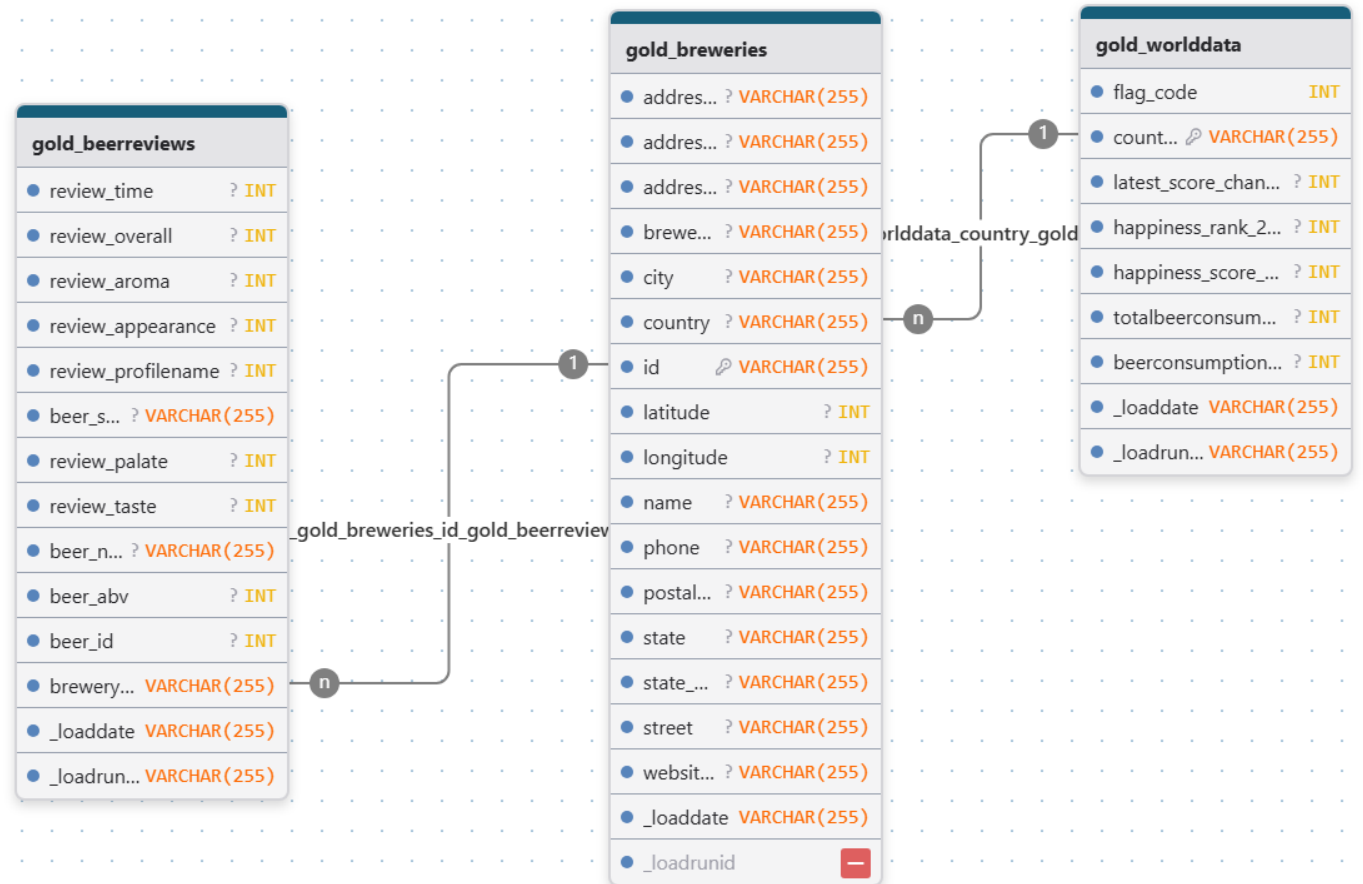
Purview

Architecture

Sources



Gold Layer



Ingest data using Fabric Data Factory, Copy Job and Mirroring





Microsoft Fabric

The unified data platform for AI transformation



Data
Factory



Analytics



Databases



Real-Time
Intelligence



Power BI



Industry
Solutions



Partner
Workloads



AI



OneLake



Purview



Data Factory in Microsoft Fabric



Enterprise data integration in Azure

- Azure Data Factory
- Synapse Link
- Purview Integrated

+



Self-service data prep with Power Query

- Excel
- Power BI
- Office 365
- Dynamics
- Power Platform

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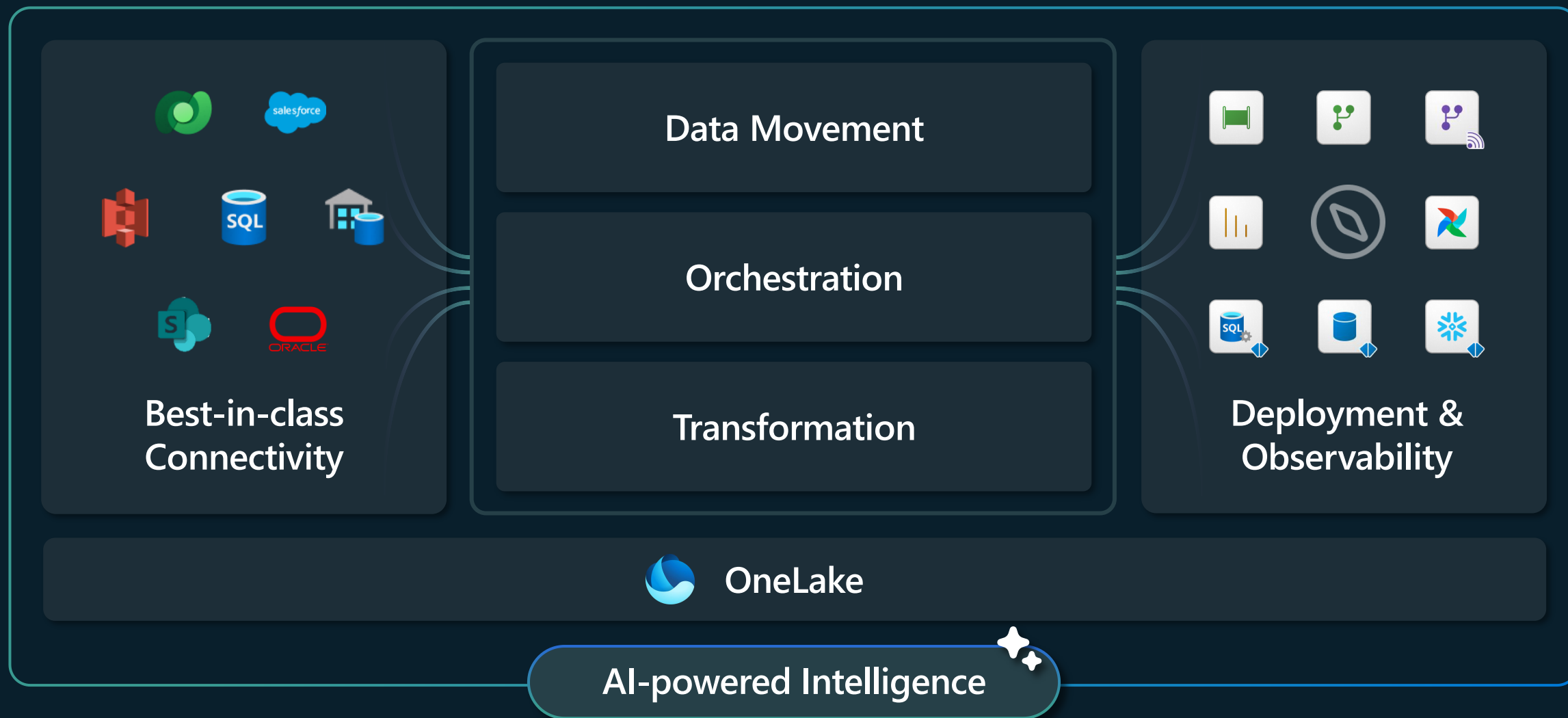
Microsoft
Fabric

Data Factory in Microsoft Fabric

- Integrated SAAS platform
- No-code & code tools
- Copilot & AI-powered
- Open Source

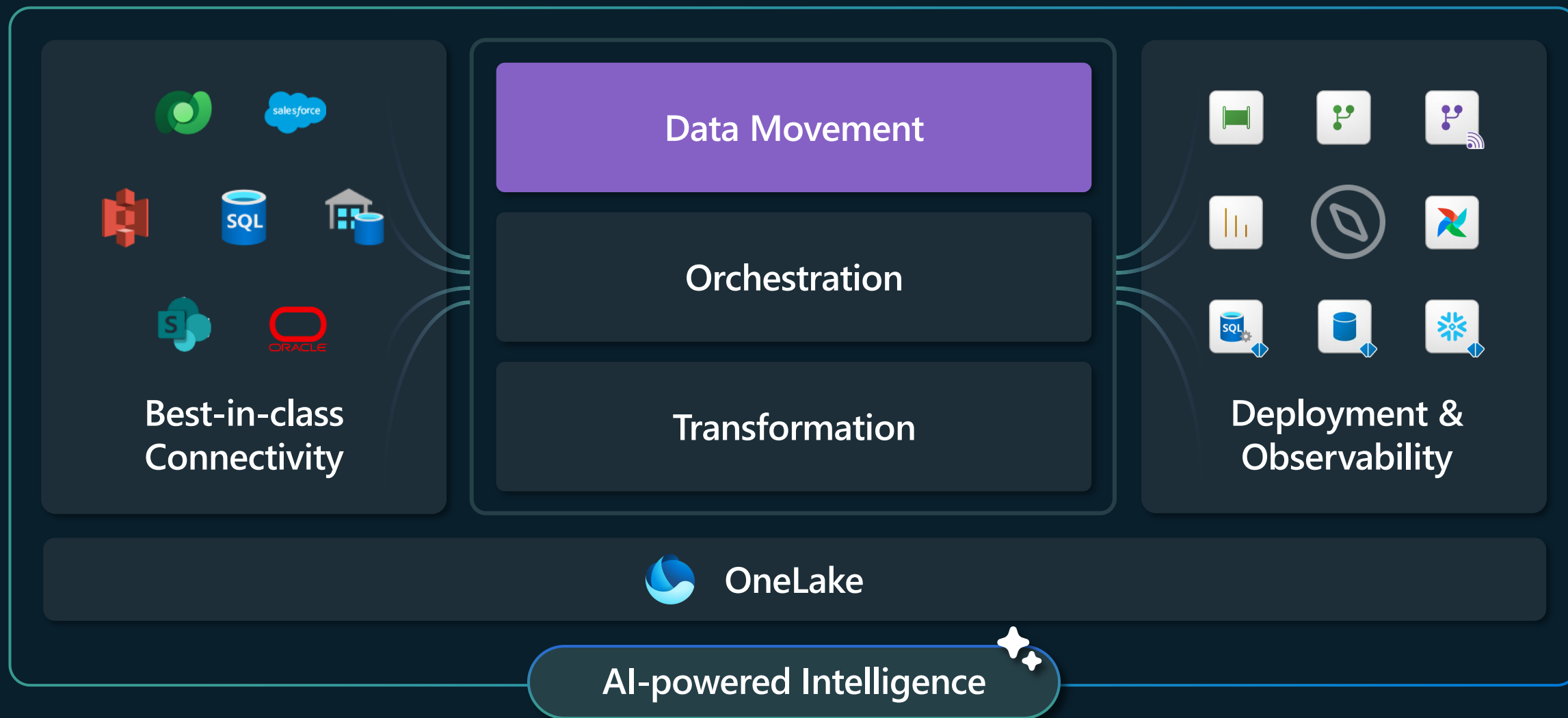


Data Factory in Microsoft Fabric





Data Factory in Microsoft Fabric





Data Factory in Microsoft Fabric

Move Data



Pipeline (Copy Activity)

Ingest data at scale and schedule data workflows.



Dataflow *Gen2*

Prep, clean, and transform data.



Mirrored Database

Easily replicate data from an existing source into Onelake.



Copy Job

Copy data easily with settings for full, incremental, or event-based.



Apache Airflow Job

Simplify creation and management of Apache Airflow environments.



Azure Data Factory

Link ADF into Fabric, and be able to use all your ADF pipelines in Fabric



Database Mirroring Deep-Dive

Unifying data in OneLake

Cross-cloud shortcuts & mirroring



First **multi-cloud**
SaaS data lake

Shortcuts for existing
data **cross clouds and**
on-premises

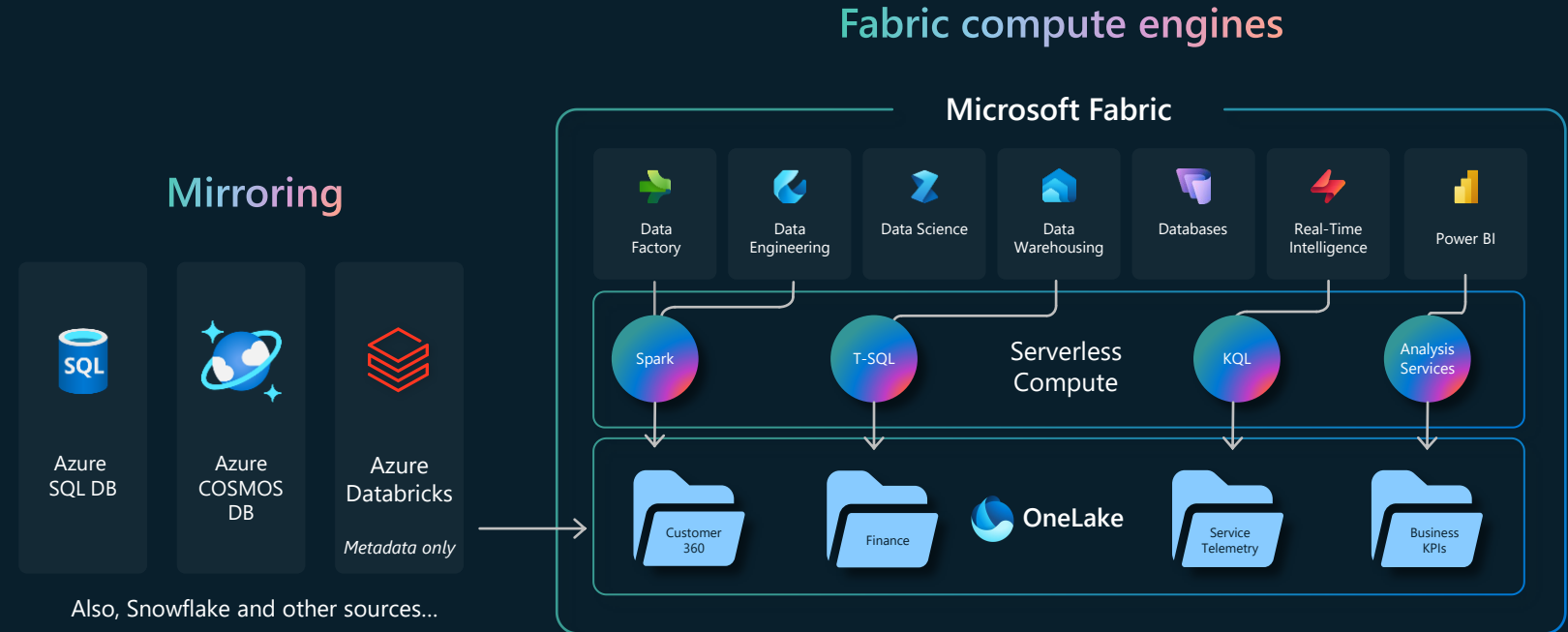
Mirroring to add
whole databases into the
OneLake data estate

Mirroring in Microsoft Fabric

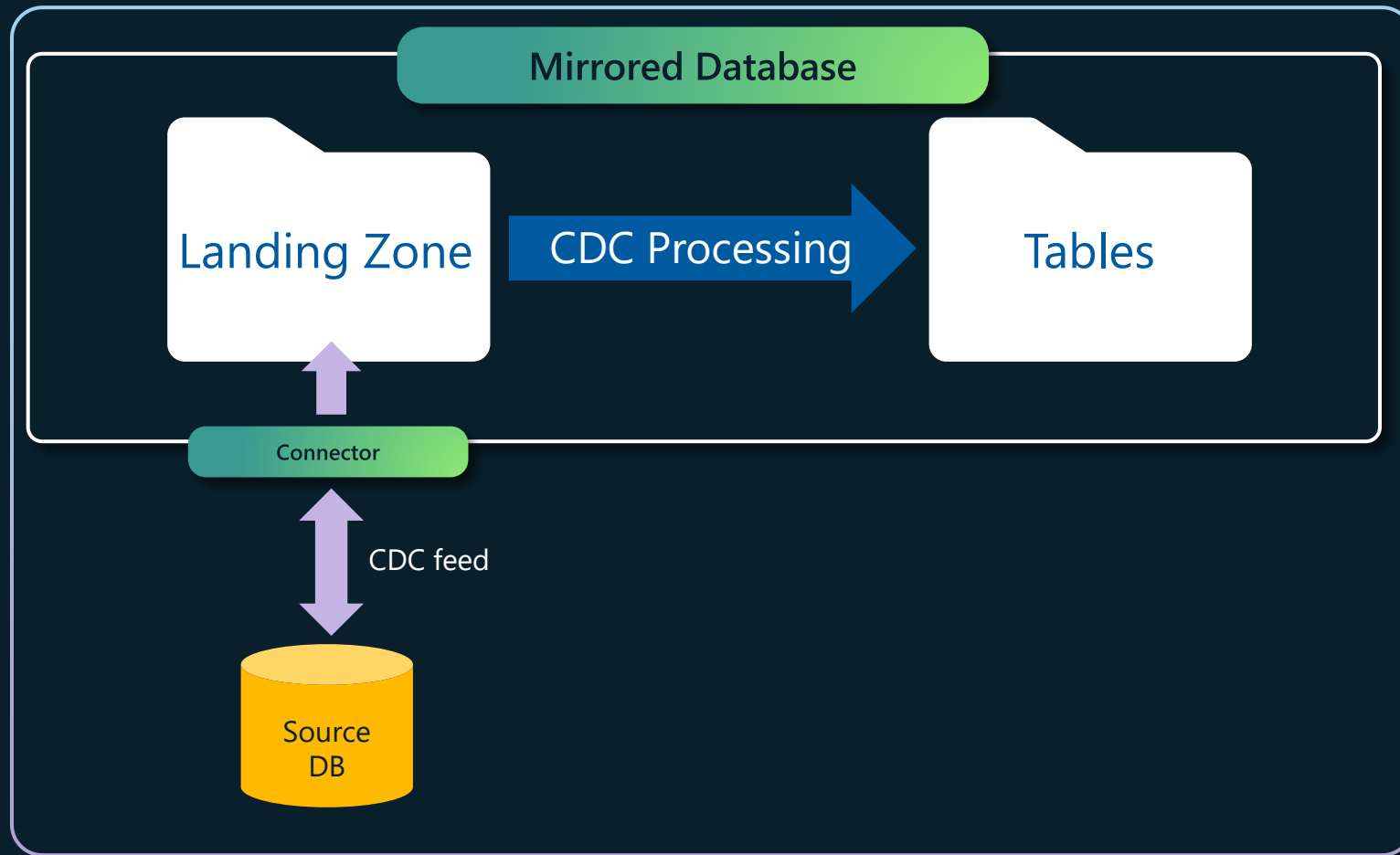
A mirrored item reflects a full external database in OneLake.

Ensures the entire database shows up in OneLake, reflecting all changes to the schema and data.

Mirroring can be done through CDC based replication for proprietary formats or by automatically maintaining a set of shortcuts to the external database for open table formats (delta and iceberg)



Database Mirroring



Mirroring allows you to add **whole databases** into the OneLake data estate

Setting up mirroring is **trivial**, the mirrored data is **automatically kept up-to-date at near-real-time**

Mirroring is **FREE***

Database Mirroring in Fabric

Automatically reflect data into OneLake

Snowflake

Mirroring for
Snowflake



Mirroring for
Azure SQL DB



Mirroring for
Cosmos DB



Mirroring for
Azure SQL MI



Mirrored Azure
Databricks Catalog

Coming Soon



Mirroring for
SQL Server



Mirroring for
Azure PostgreSQL



Mirroring for SQL
Server 2025



Mirroring for
Oracle

Free Mirroring Storage for Replicas



OneLake

Transform data using Spark and SQL



How to Transform Data

Spark

PySpark, Scala, Java, SparkSQL

Notebooks, Spark Job
Definitions, Livy API

Data Warehouse

T-SQL

Warehouse, SQL Scripts

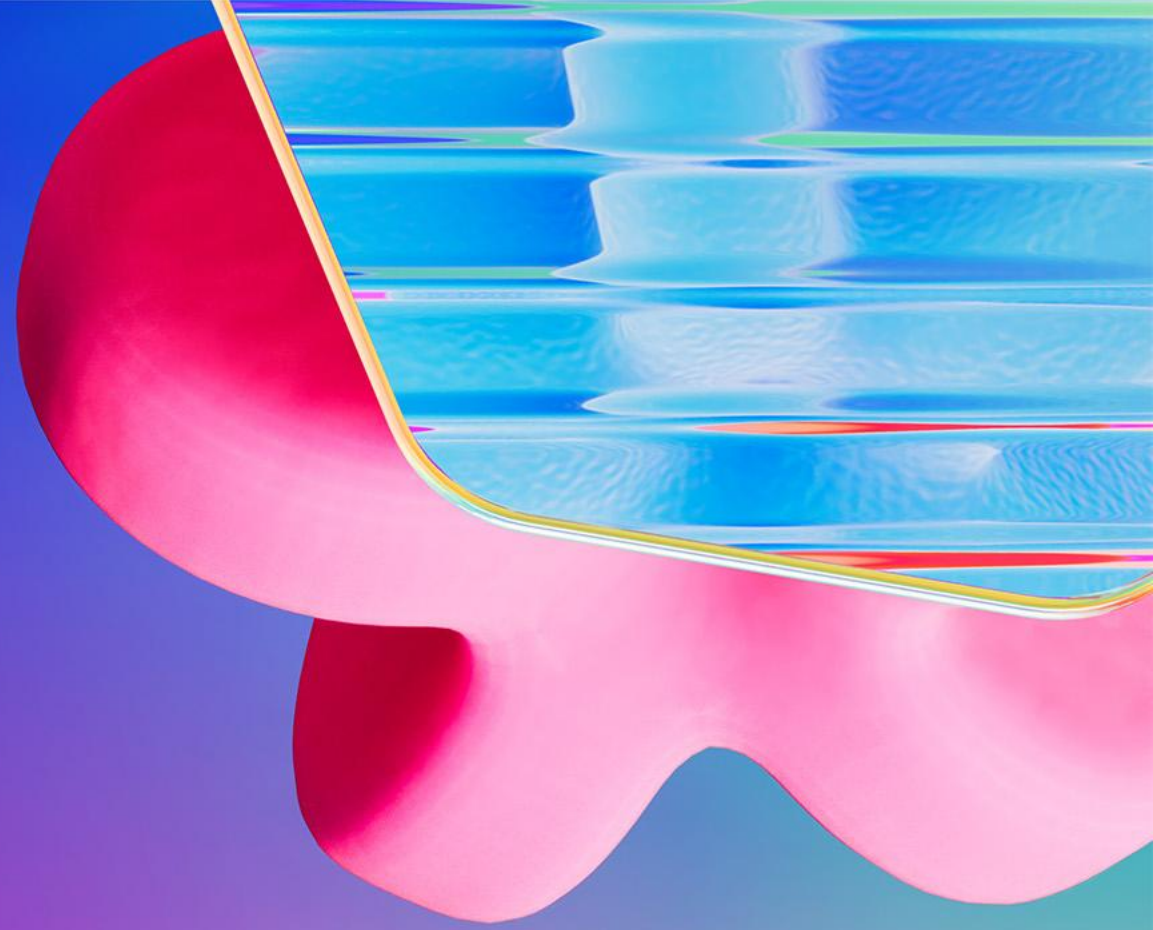
Data Flow Gen 2

M Query / Power Query

Data Flow Gen 2

Orchestrating it all....

Lunch



Query data using Fabric
Data Agents and interact
with OneLake



Foundation: OneLake unifies the data estate



Lakehouses

Databases

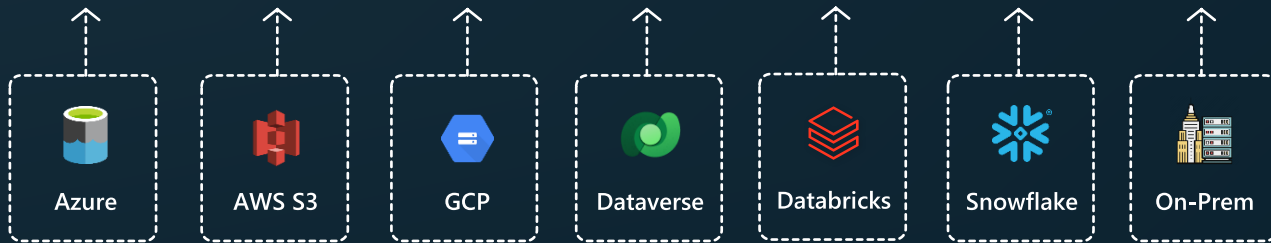
Warehouses

Eventhouses

The first step in creating a comprehensive AI-driven analytics system is to bring all the data together

OneLake unifies all data, structured and unstructured, across all clouds and databases, providing a layer of curation

OneLake provides a single pane of glass for data access, governance and security



Foundation: Define Business Semantics



Semantic Models

Lakehouses

Databases

Warehouses

Eventhouses

Semantic models are expertly curated repositories of business data and corporate metrics

Industry rush towards semantic models as key AI enablers

 Looker

 cube

 snowflake

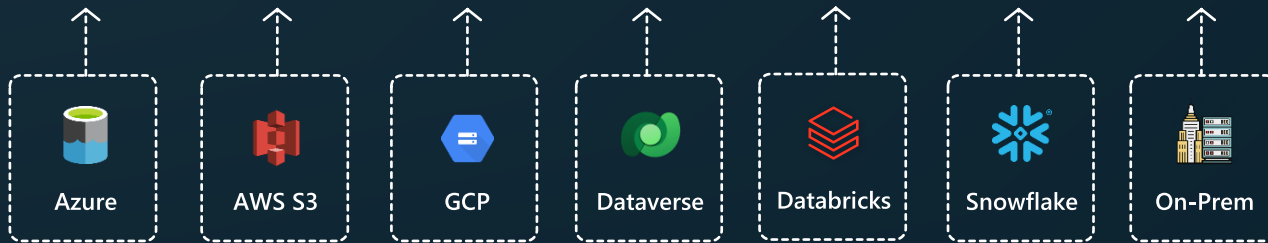
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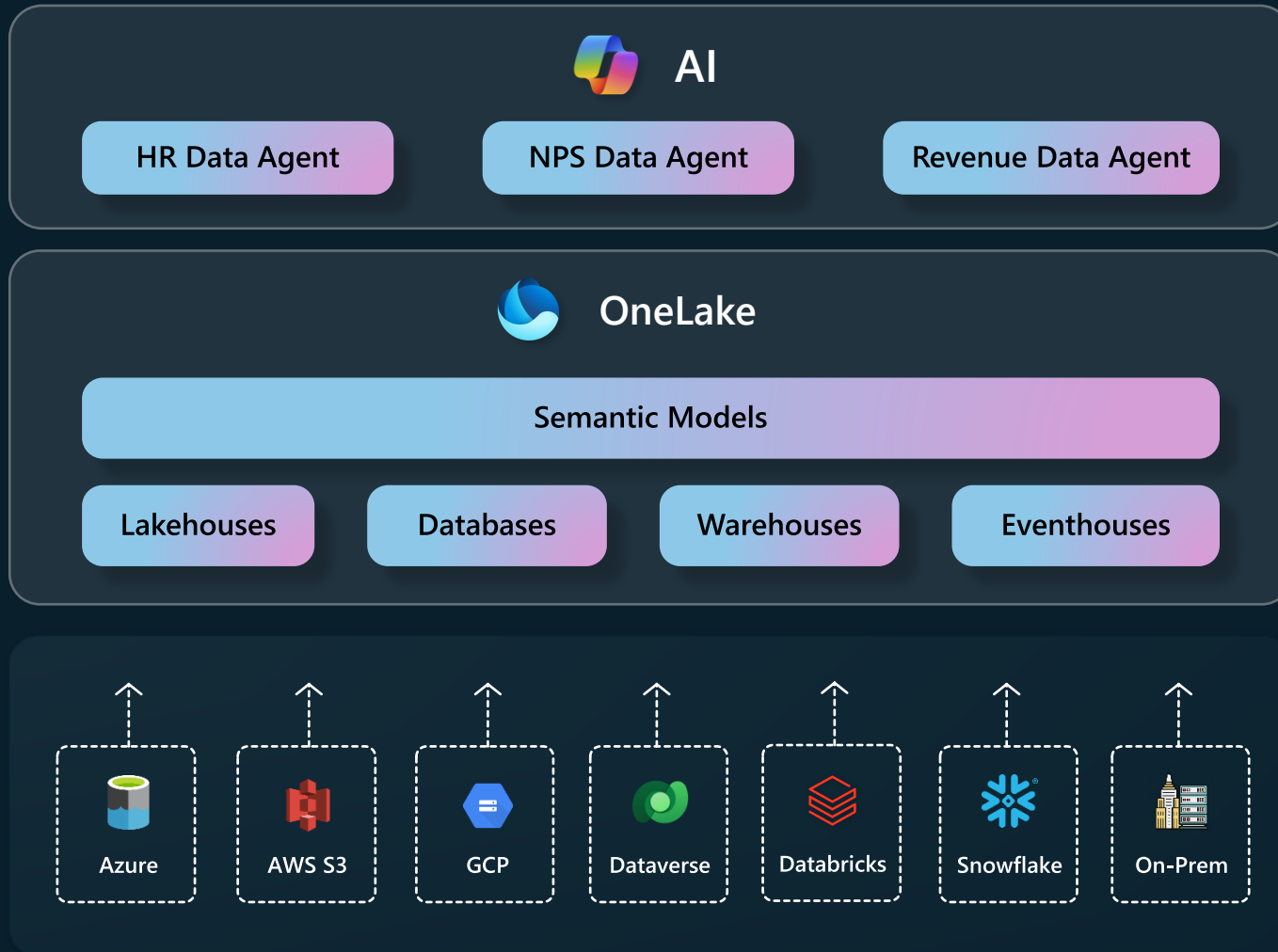
 databricks

30M MAU are regularly using 10+ million of PBI semantic models

The market footprint of PBI semantic models gives Microsoft significant leg up



Foundation: OneLake unifies the data estate



Data Agents are analytical agents that have general ability to author queries and run analysis

They are provided with a source-specific "operator's manual" on how to properly query and analyze the data

Includes instructions for when to use what data, how to understand questions, what are well known answers to common inquiries, multi-shot examples

The Data Agent can run simple queries or perform complex multi-step analysis

Tools to interact with OneLake

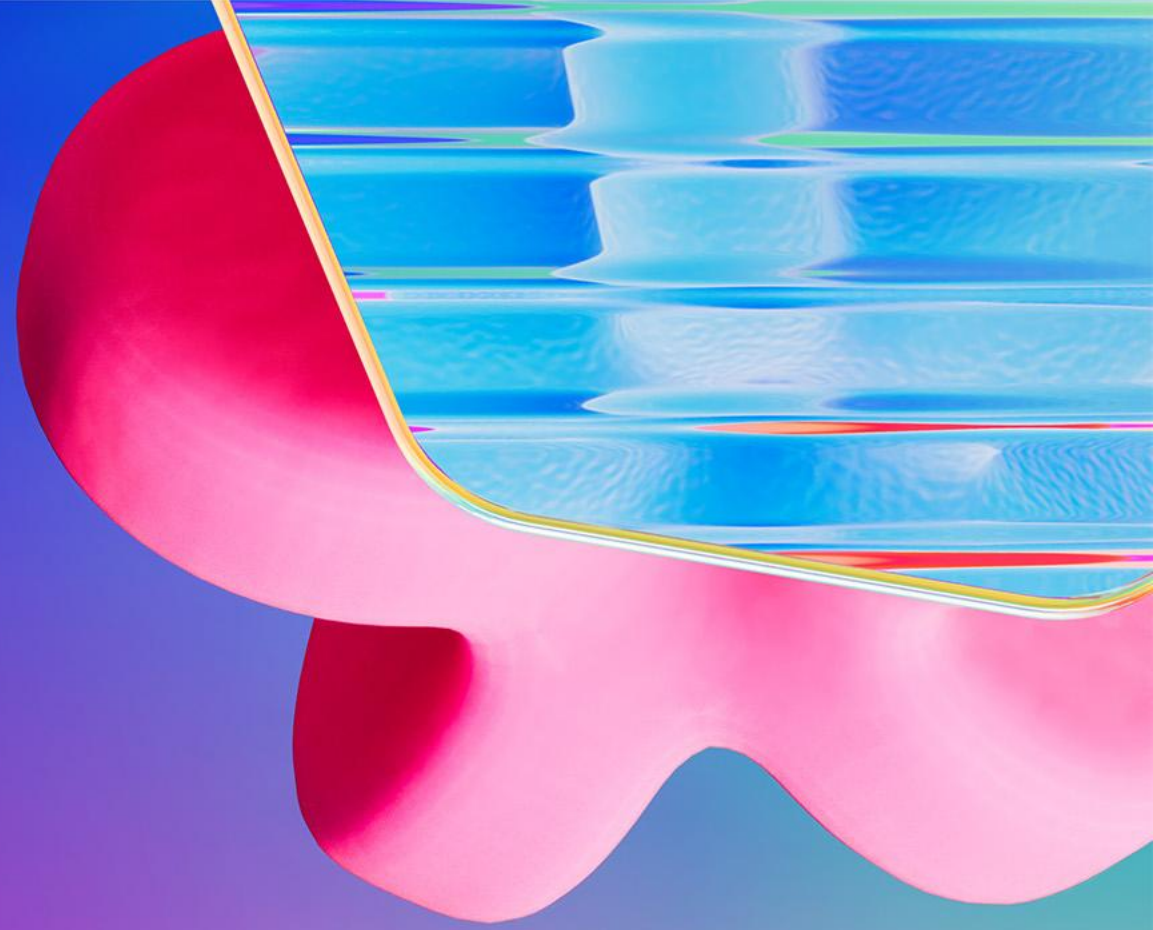


-
- Azure Storage Explorer
 - OneLake File Explorer
 - Browser
 - APIs
 - Fabric CLI
 - ...

Monitoring, admin, capacities
and automation



Wrap up and Q&A



Feedback

DataCeili 2025: Hands on with Fabric - End-to-end Data Engineering

